

Article

Fungal Presence and Changes of Wood Structure in Bark Stripping Wounds Made by Red Deer (*Cervus elaphus* L.) on Stems of *Fraxinus angustifolia* (Vahl)

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Abstract: Narrow-leaved ash (*Fraxinus angustifolia* Vahl), a highly valued European forest tree species, has been severely affected by a large-scale decline, which is most probably driven by a complex of multiple interacting factors including fungi, which contribute to and accelerate this process. Red deer (*Cervus elaphus* L.) can be considered as one of the contributing factors, as they inflict damage on the stems of young trees by stripping the bark. These wounds not only represent suitable entry points for fungi which can cause tissue necroses and decomposition, they can lead to changes in the wood structure as well. The aims of this research were to analyze chosen parameters of bark stripping wounds in narrow-leaved ash stands, identify fungi present in the tissue exposed by wounding, and inspect the effect of wounding on the wood structure. Bark stripping was observed on ash trees from 2 cm up to 18 cm of DBH and between 0.1 m and 1.9 m of stem height. The most susceptible trees were those with an average DBH of 5 ± 2.5 cm. On most of the ash trees (51%), one-third to two-thirds of the circumference was damaged. In wounded tissue, 174 fungal isolates were found, most of which belonged to known endophytic fungi from the genera *Trichoderma*, *Fusarium*, and *Clonostachys*. It was observed that earlywood cells in the wounding zone had narrower lumens compared to ones in adjacent healthy zone with regard to different trees and stem heights.

Keywords: game damage; narrow-leaved ash; *Trichoderma* spp.; *Fusarium* spp.; *Clonostachys rosea*; earlywood structure



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1. Introduction

Narrow-leaved ash (*Fraxinus angustifolia* Vahl), a widespread and important European forest trees species, has been seriously affected by a large-scale decline in recent decades. This is especially prominent in south-eastern and some parts of central Europe, where narrow-leaved ash occupies large floodplain forest areas and acts as a pioneer tree species [1,2]. An invasive pathogenic fungus, *Hymenoscyphus fraxineus*, has been identified as the main causative agent of common (*F. excelsior* L.) and narrow-leaved ash dieback in Europe; it infects ash leaves, shoots, stems, and roots, leading to wilting of leaves, loss of foliage, and formation of necrosis and cankers on all wooden plant parts [3,4]. However, more recent studies have revealed that ash decline is more probably driven by a complex of multiple interacting factors, including other fungal species, which contribute to and accelerate this process [5,6]. Ash trees are asymptotically colonized by a variety of endophytic fungi, which can become opportunistic pathogens and spread in weakened tissue, especially when the host undergoes vitality loss caused by biotic or abiotic stresses [7–11].

Rot fungi (*Armillaria* spp., *Ganoderma* spp., *Lentinus tigrinus*, etc.) are often isolated from declining ash trees, and have mostly been reported as secondary invaders which rapidly lead to root and stem deterioration and ultimately tree death and fall [9,12–14].

Although large animals are an inseparable part of forest ecosystems and capable of inflicting damage on young plants, their possible role in the decline of ash stands has not been studied very much. Research conducted in Hungary revealed no clear signs that herbivorous ungulates play a major role in the spread of *H. fraxineus* on common ash [15]. However, large ungulates such as red deer (*Cervus elaphus* L.), sika deer (*Cervus nippon* Temminck), fallow deer (*Dama dama* L.), and moose (*Alces alces* L.) are widely distributed across European forests, where they regularly cause damage to trees and stands simply by satisfying their daily and seasonal physiological needs [16]. They commonly bite green plant parts such as leaves, buds, and bark, and directly affect the structure and composition of forest vegetation [17]. Bark stripping and vegetation damage can cause economic losses [18] and enable the infection of wounded trees by various pathogenic organisms, mostly fungi [19]. Among the European large herbivores which affect forest stands, the red deer is by far most widespread [16]. There are numerous studies dealing with the analysis and different aspects of damage caused by red deer to forest stands [16,20–27]. Although deer cause bark stripping wounds on various tree species in different forest stands, in most of the studies the authors agree that debarking occurs because of a lack of fibrous food, essential nutrients, or minerals during the winter months. Several studies have focused on forest protection measures, such as providing supplemental nutrition to red deer in order to influence their feeding habits during the winter months. For example, a study conducted in Austria showed that targeted supplementary feeding in the winter months can influence the behavior of red deer and keep them in a desired area, deterring them from sensitive and endangered areas where there is a risk of stripping the bark of young trees [28]. However, the results showed effects only within a radius of 1.3 to 1.5 km of the feeding location. In contrast, Jerina et al. [29] did not establish a connection between winter feeding sites and damage to forest stands. In their research, the authors determined that microclimatic conditions are more important for the incidence of damage than the layout and average distance from the supplementary feeding place.

Nevertheless, inflicted bark stripping wounds represent suitable entry points for fungal spores and facilitate the colonization of exposed wood by various obligate or opportunistic plant pathogens and wood decay fungi, including those which are already known to contribute to ash dieback [30–32]. Additionally, endophytic fungi which were latently present in wood prior to wounding can shift to a parasitic or saprotrophic lifestyle due to changes in the substrate which favor their growth [33–35]. A study on pedunculate oak (*Quercus robur* L.) in Lithuania revealed that fungal composition in bark stripping wounds changes over time, although certain fungi can continue to grow and thrive in wounded tissue even after wound closure. The same study confirmed that younger wounds are colonized by widely distributed fungal generalists, mainly saprotrophs, while opportunistic endophytes thrived in somewhat older wounds [32]. However, this can significantly vary depending on the wounded tree species, its capability with regard to wound closure, and its overall resistance to fungal infections [36].

Additionally, despite effective sealing of wounds which forest trees usually employ, the wood structure undergoes certain changes and the timber quality deteriorates because of the discoloration and rot caused by fungi [37]. Although many external factors can affect wood structure, bark stripping is the most common one. When only phloem and cambial tissue are removed, there is a vastly different impact on tree physiology than when the xylem tissue, especially the vascular tissue, is affected [38]. A study conducted in Germany on European beech (*Fagus sylvatica* L.) showed that the area with internal wood damage was smaller than the external bark stripping area, although length of the damaged stem section was always greater inside than outside [39]. Bark stripping, which mostly occurs in young forest stands [40], affects wood quality from both the economic point of view and that of final wood production [41]. The internal damage which manifests itself in the

wooden body is of decisive importance for forestry as far as silvicultural decisions and economic profits are concerned [39]. In relation to this, additional data for quantifying the damage caused by debarking are essential. Currently, research evaluating the impacts of bark stripping is mostly directed to determining the length of time over which bark stripping occurs, the size of the damaged trees or trunks, the frequency of repeated damage, and the recovery of the resulting wounds [42].

Although the variability of wood within and among trees is perhaps the most widely studied subject in forestry [43], there is no specific model or norm in wood variation; each species can develop its own pattern in each environment and with different external effects. Cell characteristics, including cell and lumen width, can differ between the upper and lower juvenile wood of hardwoods [44]. The variability of juvenile wood properties with tree height is evident in hardwoods. Cell width plays a major role in the wood quality of hardwoods because of the vessel elements [43], with earlywood vessels having primarily a vascular role. Typically, narrow-leaved ash wood is ring-porous, with the xylem earlywood vessels being conspicuously wider in diameter in comparison to the latewood vessels. They are arranged in a ring and distributed individually or in pairs of one to four in a radial row [45]. Thus, healthy earlywood is easily distinguished from latewood in terms of vessel size [46]. In this study, damage caused by bark stripping was evaluated by quantifying one of the most important wood tissues, namely, the vascular tissue.

In light of the above, the aims of this research were to: (1) measure the intensity of stem wounding inflicted by red deer on *F. angustifolia* stems; (2) isolate and identify fungi present in the wounded tissue; and (3) compare the wood structures of wounded and adjacent healthy tissue in order to gain a first insight into the contribution of red deer damage to the complex process of ash decline.

2. Materials and Methods

2.1. Field Measurement and Sampling

Research was conducted in lowland forests at the Sava river basin, where narrow-leaved ash occurs naturally and occupies the largest continuous forest area (28,000 ha) in Croatia [47]. Measurements of bark stripping wounds on stems were carried out in 2021 and 2022 in three relatively distant *F. angustifolia* forest stands less than 15 years old that were known to be in the area inhabited by red deer (Figure 1, Table 1). For this purpose, 20 m wide transects were laid out in each forest stand at least 20 m away from the forest edge. Transects were evenly distributed across the stand surface area in such a way that their total area represented 10% of the total forest stand surface area; the length and the number of transects in each forest stand was varied accordingly. All present trees were recorded on each transect, and the following features were noted or measured for each:

- tree species;
- diameter at breast height (DBH, cm), measured at the 1.30 m mark of stem height;
- presence of red deer damage on stem (visible bark stripping, yes/no) (Figure 2);
- bark stripping intensity (1—less than 1/3 of circumference damaged, 2—1/3 to 2/3 of circumference damaged, 3—more than 2/3 of circumference damaged);
- height at which bark stripping occurred (the lowest and the highest point on each stem, m);
- total length of visible bark stripping on a stem (m).

In location L1, ten samples of damaged narrow-leaved ash stems approximately 25 cm in length were randomly collected for fungal analysis and stored at 4 °C for not more than 24 h before further processing. The sampled stems reflected different wounding stages, mostly closed wounds with formed callus. For the analysis of changes in wood structure, 5 cm thick cross-sections of narrow-leaved ash stem were taken from older trees (approximately 20 years old) growing in the vicinity of location L1. Five trees were sampled from that location, and cross-section disks were taken from four different stem heights on each tree (0.25, 0.75, 1.25 and 1.75 m).

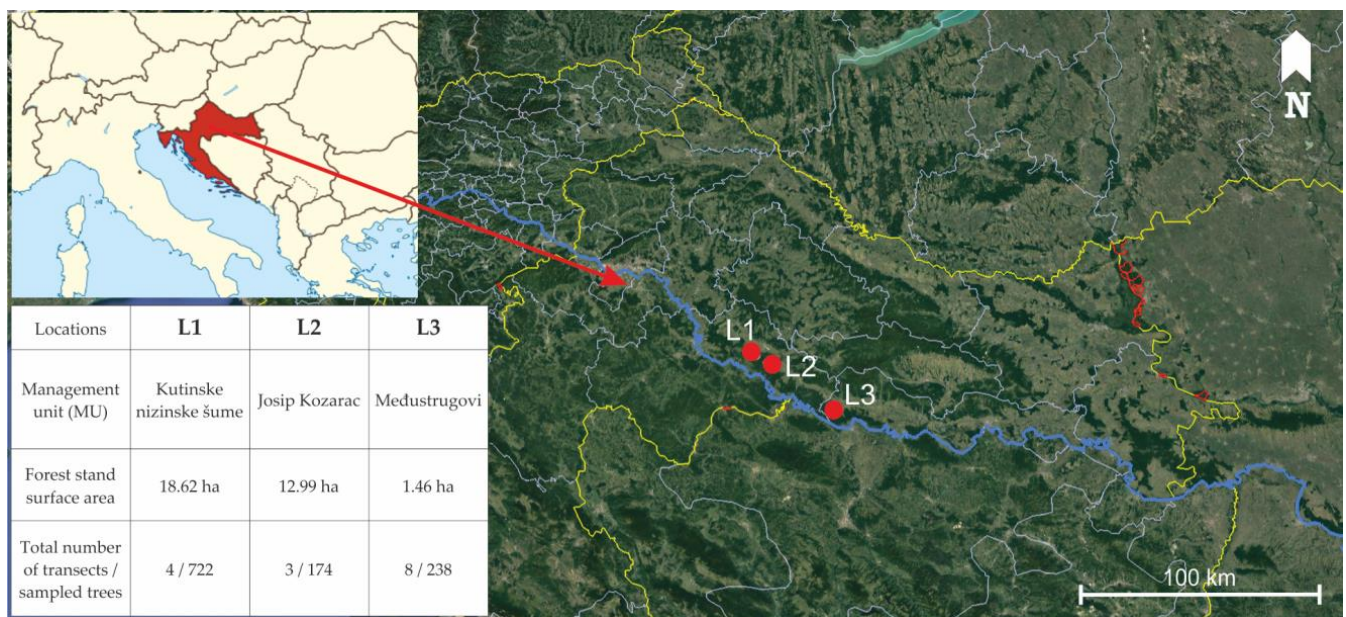


Figure 1. Locations of three narrow-leaved ash forest stands in the Sava river basin where bark stripping wounds made by red deer were studied.

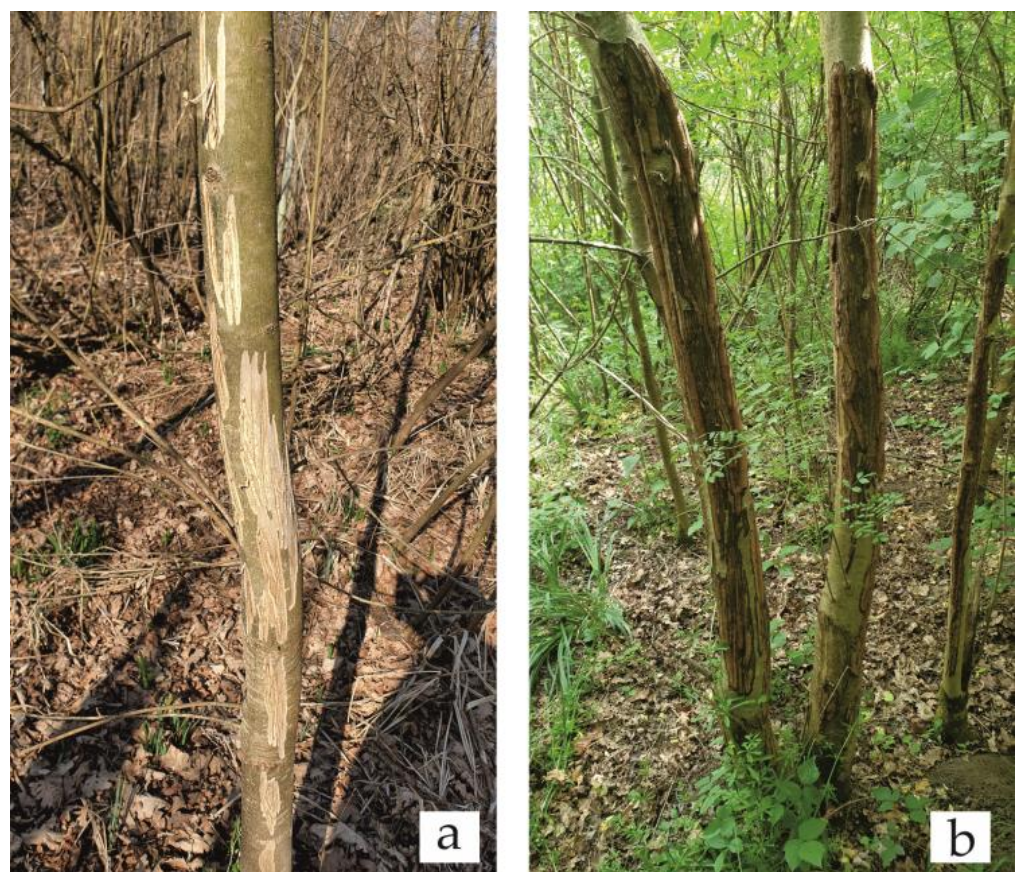


Figure 2. Examples of bark stripping wounds inflicted by red deer on narrow-leaved ash: (a) open wounds without visible formation of callus and (b) partially or fully closed wounds with visible callus formation.

Table 1. Habitat, climate, and stand characteristics of three narrow-leaved ash forest stands in the Sava river basin where bark stripping wounds made by red deer were studied.

	L1	L2	L3
Coordinates	45.431646 N, 16.735151 E	45.403425 N, 16.787225 E	45.203113 N, 17.152394 E
Elevation (above sea level, m)	94–95	94–95	91–92
Slope (°)	0	0	0
Phytocenosis	forest of narrow-leaved ash with autumn snowflake (As. <i>Leucojo-Fraxinetum angustifoliae</i> Glavač 1959)	forest of pedunculate oak with <i>Genista elata</i> (As. <i>Genisto elatae-Quercetum roboris</i> Ht. 1938)	forest of narrow-leaved ash with autumn snowflake (As. <i>Leucojo-Fraxinetum angustifoliae</i> Glavač 1959)
Forest type	regular, even aged forest of narrow-leaved ash	regular, even aged forest of pedunculate oak with narrow-leaved ash	regular, even aged forest of narrow-leaved ash
Average tree age	<15 years	<15 years	<15 years
Density (No. of trees/ha)	388	134	1698
The average DBH of narrow-leaved ash trees (cm)	10	7	5
Soil type	amphigley	amphigley	amphigley
Mean annual temperature (°C)	11.01	11.01	11.1
Mean temperature in the vegetation period (°C)	17.6	17.6	17.7
Mean annual precipitation (mm)	861.3	861.3	770.3
Mean precipitation in the vegetation period (mm)	506.7	506.7	441.0
Understory vegetation	shrubs rarely present, ground vegetation mostly presented by <i>Leucojum aestivum</i>	rich shrub layer mostly presented by <i>Genista elata</i>	shrubs rarely present, ground vegetation mostly presented by <i>Leucojum aestivum</i>

2.2. Isolation and Identification of Fungi

From each stem sample, 0.5 cm thick cross-sections with visible wood or bark discoloration or necrosis were taken and surface-sterilized by immersion in 70% ethanol for one min followed by rinsing in sterile distilled water and drying on sterile filter paper under a laminar flow hood (Iskra Pio, Šentjernej, Slovenia). Xylem subsamples (wood chips, 5 × 5 mm) were then aseptically taken at the edge of discoloration or necrosis and plated on malt extract agar (MEA, Oxoid, Basingstoke, UK) supplemented with streptomycin sulphate (200 mg/L, Sigma-Aldrich, St. Louis, MI, USA). Petri dishes were incubated in the dark at 19–20 °C for four weeks and checked daily for fungal growth. Emerging mycelia were transferred onto fresh MEA to obtain pure fungal cultures. Based on their morphological characteristics, the obtained isolates were grouped into putative species, that is, morphotypes, and least one representative isolate from each morphotype was chosen for identification using molecular methods.

DNA was extracted from mycelia cultured in malt extract broth (MEB, Liofilchem, Roseto degli Abruzzi, Italy) in 2 mL microtubes for 7 days by the salting-out method according to Ceniz [48]. Minor modifications were made to the composition of the extraction buffer (20 mM Tris HCl, 200 mM NaCl, 2 mM EDTA, 10% SDS, pH 8.0) and to the homogenization procedure, where the mycelium pellet was subjected to grinding in a Tissue Lyser II (Qiagen, Hilden, Germany) for 3 min at 30 Hz with addition of a single 5 mm stainless steel bead per microtube, rather than crushing it with a conical grinder. In Polymerase Chain Reaction (PCR), the internal transcribed spacer (ITS) region of the ribosomal DNA (rDNA) was amplified using primers ITS1-F [49] and ITS 4 [50]. PCR was carried out in 65 µL reactions containing 200 µM deoxyribonucleoside triphosphates, 0.2 µM of each primer, 1.5 U of Taq DNA polymerase with 1 × reaction buffer (Sigma-Aldrich, St. Louis,

MO, USA), 2.75 mM MgCl₂, and 1 µL of 100-fold diluted DNA template. The cycling conditions were as follows: an initial denaturation at 95 °C for 5 min, 35 cycles of denaturation at 95 °C for 30 s, annealing at 55 °C for 30 s, extension at 72 °C for 30 s, and a final extension step at 72 °C for 7 min.

The resulting PCR products were sequenced using the same primers as in the PCR at the DNA sequencing facility of MacroGen Europe (Amsterdam, The Netherlands). After processing raw data using BioEdit Sequence Alignment Editor v.7.2.5 software [51], sequences were identified by comparison with reference sequences in National Center for Biotechnology Information (NCBI) GenBank using the Basic Local Alignment Search Tool (BLAST + 2.15.0) [52]. Sequences with 99.0%–100% similarity were identified to the species level and with 95.0%–98.99% similarity to the genus level.

2.3. Sample Preparation and Wood Structure Analysis

Cross-section disks taken from each of the four stem heights were naturally dried and finely sanded (granulation 320) in order to make visible the wounds within the annual growth rings. Disks were first visually observed, and the diameter was measured. According to wound presence, the same single annual growth ring from the pith was selected and marked along one radius of each disk: two zones by type, including one with the wound present and a control zone (a healthy intact part of the annual growth ring).

Each selected annual growth ring was cut in the form of two sample blocks sized approximately 10 (tangential) × 10 (radial) × 20 (longitudinal) mm. Sample preparation included softening using boiled distilled water for 120 min and cutting into transverse sections (30 µm thick) using a sliding microtome (A.C. Reichert, Vienna, Austria). The sections were stained with a mixture of Safranin (ThermoFisher (Kandel) GmbH, Kandel, Germany) and Astra blue (Carl Roth GmbH&Co KG, Karlsruhe, Germany) for 15 min, washed multiple times in a 70% and 96% ethanol solution until the stain was completely rinsed, and mounted on microscope slides using Euparal (Carl Roth GmbH&Co KG, Karlsruhe, Germany). Four images were taken as consecutive replicates from the earlywood region in the transverse section using a light microscope (Carl Zeiss Jena, Germany) and DinoCapture 2.0 software (Dino-Lite Europe, The Netherlands) connected to a 5 MP digital camera (Dino-Lite Europe, The Netherlands). The samples were photographed at ×60 magnification. Earlywood vessel lumen diameter (EVLD) (µm) measurements were performed using ImageJ software v. 1.8.0 [53].

2.4. Statistical Analysis

Differences in the DBH of the analyzed trees among different locations and categories of bark stripping intensity were tested using analysis of variance (One-way ANOVA). Chi-square tests of independence were used to test the significance of the association between the presence of red deer wounding on narrow-leaved ash stems and the researched location. Pearson correlation values were calculated between the DBH of wounded narrow-leaved ash trees and the measured parameters of inflicted bark stripping wounds (height at which bark stripping occurs, total length of visible bark stripping on a stem).

Regarding the wood structure analysis, the descriptive statistics (mean and standard deviation) were calculated for the EVLD of two zones at different stem heights on all five trees. Repeated measures analysis of variance (ANOVA) was used to test the effect of the stem height, tree, wood zone, and their interactions. This analysis was used to determine the significance of differences in EVLD in several instances: in trees between two zones and four stem heights, between zones of four stem heights, and in the case of all three parameters together. Further analysis was based on finding differences in the EVLD of two zones by observing four different stem heights separately.

In all performed tests, a *p*-value less than 0.05 was considered statistically significant. Statistical analyses were performed in the TIBCO Statistica 13.5.0 software package (TIBCO Software Inc., Palo Alto, CA, USA).

3. Results

3.1. Inventory of Bark Stripping in the Analyzed Forest Stands

In total, 1134 trees were observed and measured in three locations. Most of the trees belonged to narrow-leaved ash (78.3%). Other observed species were pedunculate oak (14.5%), elm (*Ulmus* sp., 5.9%), black alder (*Alnus glutinosa*, 0.9%), and wild pear (*Pyrus pyraeaster*, 0.4%). More than half of ash trees (56%) displayed bark stripping wounds, whereas this was visible on only 2% of oak and 9% of elm trees, and was not noticed on black alder or wild pear (Figure 3).

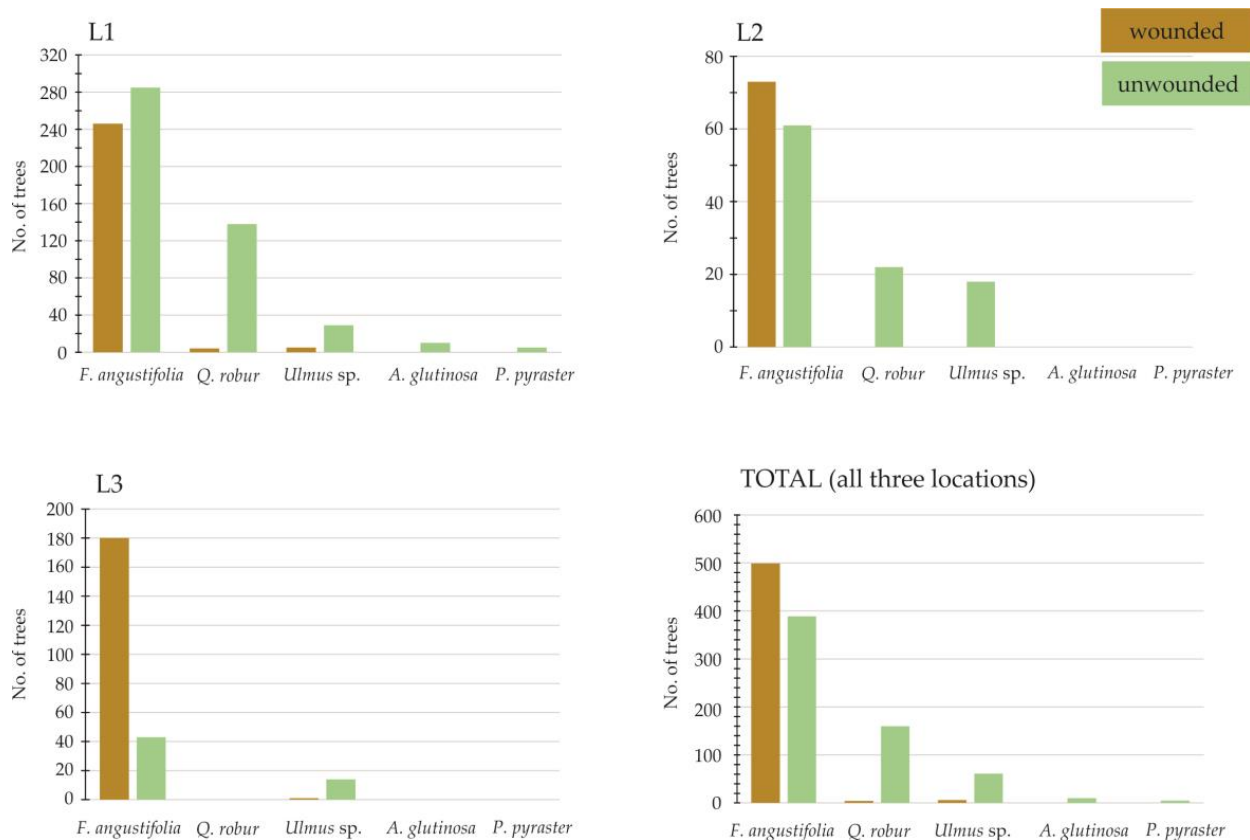


Figure 3. Number of wounded trees (in dark brown) and unwounded trees (in green) distributed by species and location.

The average DBH of narrow-leaved ash trees (Table 1) differed significantly among the three locations ($F(2, 885) = 62.81, p < 0.001$), as well as in the presence of bark stripping wounds on stems ($\chi^2(2) = 75.64, p < 0.001$).

The average DBH of the wounded narrow-leaved ash trees in all three locations was 5 cm (SD = 2.5 cm). Bark stripping was observed on ash trees from 2 cm up to 18 cm of DBH, between 0.1 m and 1.9 m of stem height (average bark stripping height = 0.6 ± 0.4). The average length of bark stripping wounds was 0.6 m (SD = 0.3 m). This varied somewhat among the researched locations (Figure 4, Table 2).

Table 2. The average height at which bark stripping occurred and average length of visible bark stripping on stems of wounded narrow-leaved ash trees in the three researched locations.

	L1	L2	L3
The average lowest point of bark stripping ($\pm$ SD) (m)	0.4 ($\pm$ 0.2)	0.5 ($\pm$ 0.3)	0.3 ($\pm$ 0.2)
The average highest point of bark stripping ($\pm$ SD) (m)	1.0 ($\pm$ 0.3)	1.0 ($\pm$ 0.3)	0.9 ($\pm$ 0.2)
The average length of visible bark stripping ($\pm$ SD) (m)	0.6 ($\pm$ 0.3)	0.5 ($\pm$ 0.2)	0.6 ($\pm$ 0.2)

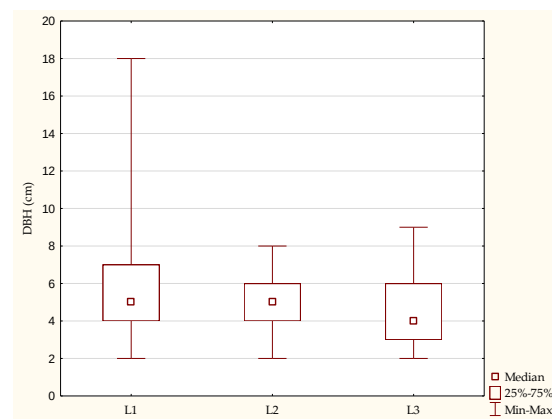


Figure 4. DBH of the wounded narrow-leaved ash trees in the three researched locations.

On most of the ash trees (51%), one-third to two-thirds of the circumference was damaged (Figure 5).

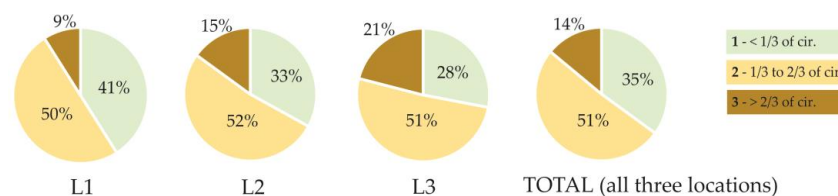


Figure 5. Share of wounded narrow-leaved ash trees with different bark stripping intensity (1—less than 1/3 of circumference damaged, 2—1/3 to 2/3 of circumference damaged, 3—more than 2/3 of circumference damaged) in the three researched locations and in total.

According to ANOVA, the DBH of wounded ash trees did not significantly vary among different categories of bark stripping intensity ($F(2, 496) = 0.01$, $p = 0.98$), indicating that these parameters are not significantly associated. Weak positive correlations were found between the height at which bark stripping occurred (the lowest and the highest point on each stem) and the DBH of wounded trees (Table 3). The height of wounds on the stem increased with the increase in DBH as well.

Table 3. Pearson correlation values between DBH of wounded narrow-leaved ash trees and measured parameters of inflicted bark stripping wounds (red-marked correlations are significant at $p < 0.05$).

Variable	DBH	Lowest Point of Bark Stripping on a Stem	Highest Point of Bark Stripping on a Stem	Length of Bark Stripping on a Stem
DBH	1.000	0.181	0.153	0.004
Lowest point of bark stripping on a stem	0.181	1.000	0.441	−0.394
Highest point of bark stripping on a stem	0.153	0.441	1.000	0.651
Length of bark stripping on a stem	0.004	−0.394	0.651	1.000

3.2. Fungi Present in Bark Stripping Wounds

The sampled wounded narrow-leaved ash stems reflected various stages of symptom development following bark stripping, from freshly exposed wood with only phloem discoloration visible on the cross-section to partly closed (partially formed callus) or fully closed (completely formed callus) cankers with specific limited discolorations or rot visible on cross-sections, which mostly followed the line of individual tree rings or were placed in the central part of the stem (often of a triangular shape) (Figure 6).

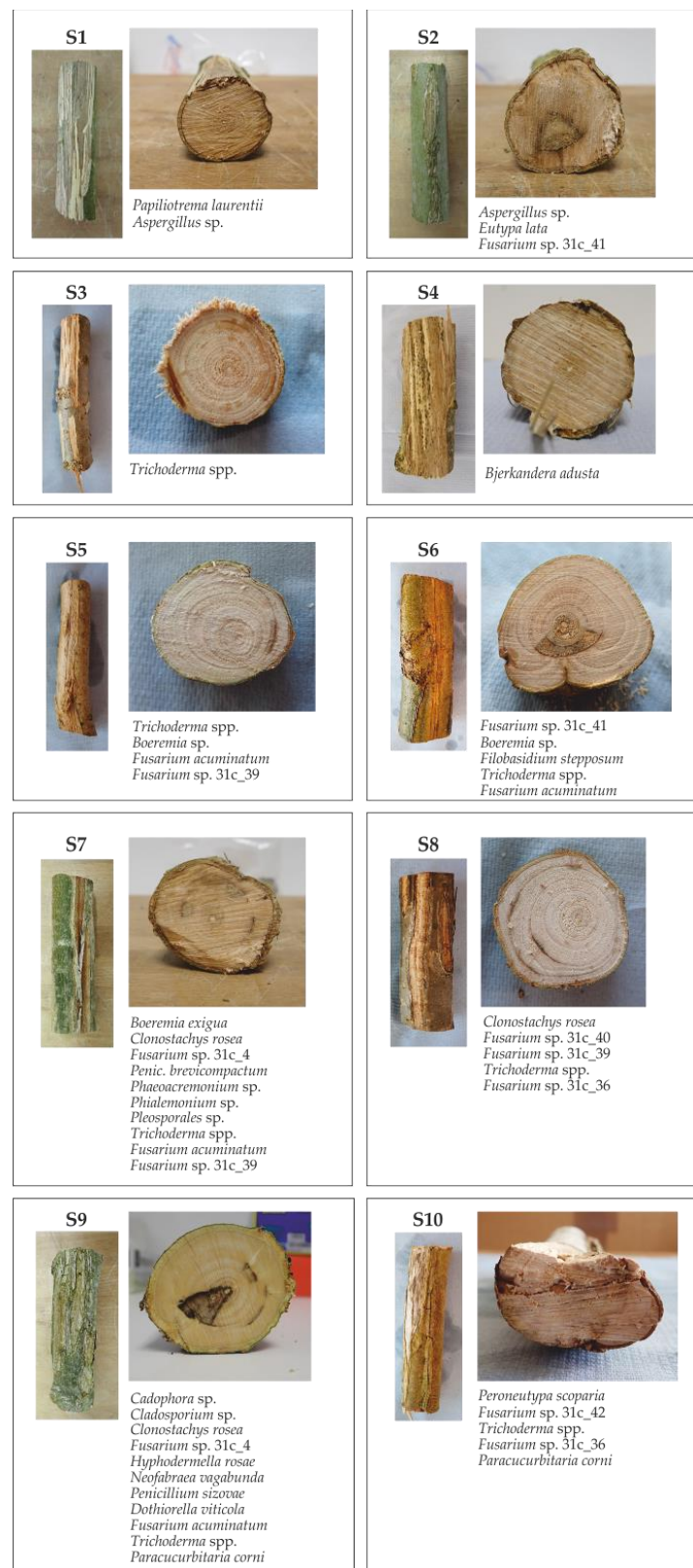


Figure 6. Sampled wounded narrow-leaved ash stems reflecting various stages of symptom development following bark stripping, with lists of associated fungi isolated in this study (Stem samples: S1, S3, and S4—freshly exposed wood with only phloem discoloration; S2, S6, S7, and S9—closed wound with triangular shape discoloration and/or rot on a cross-section; S5 and S8—closed wound with a discoloration limited to the line of an individual tree ring on a cross-section; S10—partially closed wound with irregularly shaped discoloration and rot on a cross-section).

From of ten wounded stem samples, 236 xylem subsamples (wooden chips) were plated on agar media, resulting in growth of 174 isolates that represented 28 fungal taxa (Table 4), of which 24 were members of the phylum Ascomycota. Of the 28 taxa, 14 were identified to species level, 13 taxa to genus level, and one taxon to order level. At least one fungal taxon was isolated from each stem, with an average number of five different taxa per stem. The maximum number of taxa found on one stem was 11. The most frequently isolated were various species of genus *Trichoderma*, which were combined together and referred to as *Trichoderma* spp., due to difficult delimitation and identification of species in this genus and limited lab resources. Other most common species found in bark stripping wounds were *Clonostachys rosea* and *Fusarium acuminatum*, as well as several other *Fusarium* species which remained identified only to a genus level. *Eutypa lata* was highly abundant regarding the number of obtained isolates (29), but was found in only one stem.

Table 4. Number of obtained isolates and colonized stems by fungal taxa found in bark stripping wounds inflicted by red deer on narrow-leaved ash.

Phylum	Fungal Taxon	GenBank Accession No.	No. of Obtained Isolates	No. of Colonized Stems
A	<i>Aspergillus</i> sp.	OR808055	3	2
A	<i>Boeremia exigua</i>	OR808056	1	1
A	<i>Boeremia</i> sp.	OR808057	5	2
A	<i>Cadophora</i> sp.	OR808058	1	1
A	<i>Cladosporium</i> sp.	OR808059	1	1
A	<i>Clonostachys rosea</i>	OR808060	17	3
A	<i>Dothiorella viticola</i>	OR808061	1	1
A	<i>Eutypa lata</i>	OR808062	29	1
A	<i>Fusarium acuminatum</i>	OR808063	13	4
A	<i>Fusarium</i> sp. 31c_36	OR808064	3	2
A	<i>Fusarium</i> sp. 31c_42	OR808065	2	2
A	<i>Fusarium</i> sp. 31c_4	OR808066	12	2
A	<i>Fusarium</i> sp. 31c_41	OR808067	2	2
A	<i>Fusarium</i> sp. 31c_40	OR808068	1	1
A	<i>Fusarium</i> sp. 31c_39	OR808069	12	3
A	<i>Neofabraea vagabunda</i>	OR808070	2	1
A	<i>Paracucurbitaria corni</i>	OR808071	2	2
A	<i>Penicillium brevicompactum</i>	OR808072	2	1
A	<i>Penicillium sizovae</i>	OR808073	2	1
A	<i>Peroneutypa scoparia</i>	OR808074	6	1
A	<i>Phaeoacremonium</i> sp.	OR808075	3	1
A	<i>Phialemonium</i> sp.	OR808076	12	1
A	<i>Pleosporales</i> sp.	OR808077	2	1
A	<i>Trichoderma</i> spp.	–	32	5
B	<i>Bjerkandera adusta</i>	OR808078	1	1
B	<i>Filobasidium stepposum</i>	OR808079	2	1
B	<i>Hyphodermella rosae</i>	OR808080	4	1
B	<i>Papiliotrema laurentii</i>	OR808081	1	1

3.3. Differences in Wood Structure in Relation to Zones, Trees, and Stem Height

The stem diameter of all five trees was measured at 0.25, 0.75, 1.25, and 1.75 m of the tree height (Table 5).

Table 5. The mean and standard deviation of narrow-leaved ash earlywood vessel lumen diameter (EVLD) of the control (healthy) and wounded zones and the stem diameter (D) from four different stem heights.

Tree	Property	Data	Type	Stem Height (m)				
				0.25	0.75	1.25	1.75	Total (All Heights)
1	EVLD (μm)	MEAN $\pm$ SD	C	151.24 $\pm$ 36.53	183.09 $\pm$ 33.02	171.20 $\pm$ 35.50	238.93 $\pm$ 41.98	186.12 $\pm$ 36.76
			D	148.38 $\pm$ 27.92	80.63 $\pm$ 25.92	97.25 $\pm$ 23.88	119.71 $\pm$ 64.41	111.49 $\pm$ 35.53
	D (cm)	MEAN	S	35.00	31.00	29.00	22.00	29.25
2	EVLD (μm)	MEAN $\pm$ SD	C	168.43 $\pm$ 43.83	118.14 $\pm$ 34.67	174.50 $\pm$ 35.21	141.31 $\pm$ 29.15	150.60 $\pm$ 35.72
			D	91.90 $\pm$ 20.86	72.54 $\pm$ 24.75	88.83 $\pm$ 30.54	120.49 $\pm$ 32.41	93.44 $\pm$ 27.14
	D (cm)	MEAN	S	24.00	21.00	19.50	16.50	20.25
3	EVLD (μm)	MEAN $\pm$ SD	C	177.60 $\pm$ 28.67	140.91 $\pm$ 33.95	159.54 $\pm$ 29.50	176.62 $\pm$ 24.96	163.67 $\pm$ 29.27
			D	162.91 $\pm$ 35.36	101.29 $\pm$ 32.18	126.86 $\pm$ 28.60	88.94 $\pm$ 23.12	120.00 $\pm$ 29.82
	D (cm)	MEAN	S	27.00	22.00	20.50	17.50	21.75
4	EVLD (μm)	MEAN $\pm$ SD	C	151.96 $\pm$ 35.98	178.13 $\pm$ 27.38	168.26 $\pm$ 36.34	235.87 $\pm$ 39.29	183.56 $\pm$ 34.75
			D	147.81 $\pm$ 29.65	81.46 $\pm$ 23.04	99.41 $\pm$ 19.22	118.39 $\pm$ 54.14	111.77 $\pm$ 31.51
	D (cm)	MEAN	S	33.50	30.50	29.00	24.00	29.25
5	EVLD (μm)	MEAN $\pm$ SD	C	167.34 $\pm$ 37.15	120.81 $\pm$ 29.97	172.54 $\pm$ 33.83	141.81 $\pm$ 28.15	150.63 $\pm$ 32.28
			D	87.78 $\pm$ 18.91	71.42 $\pm$ 20.15	84.04 $\pm$ 29.21	119.25 $\pm$ 29.92	90.62 $\pm$ 24.55
	D (cm)	MEAN	S	25.50	22.50	19.00	16.50	20.88

Notes: EVLD—earlywood vessel lumen diameter; D—stem diameter; C—control (healthy) zone; S—stem; SD—standard deviation.

In total, 20 disks were sampled at four different stem heights, and 40 histological preparations with cross-sections (20 of control (healthy) zone and 20 of wounded zone) were used in measurements. The mean values and standard deviations for the EVLD of the control (healthy) and wounded zones from all five trees are provided in Table 5. The trends in the EVLD of the control (healthy) and wounded zones with height are shown in Figure 7.

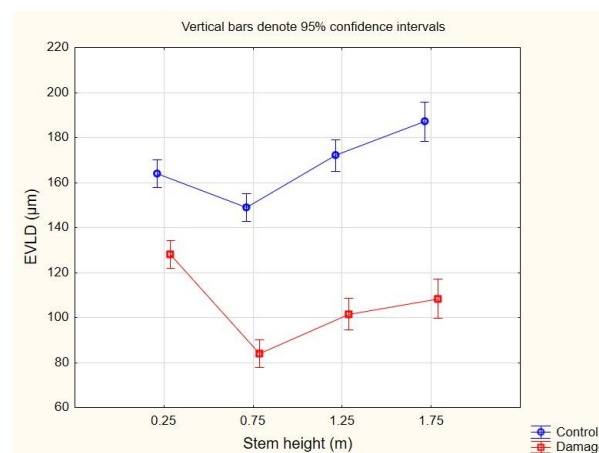


Figure 7. Distribution of earlywood vessel lumen diameter (EVLD) in the control (healthy) and wounded zones by stem height (vertical bars denote 95% confidence intervals).

The lowest mean EVLD was measured in the control (healthy) zone at a stem height of 0.25 m in trees 1 and 4 and at stem height 0.75 m in trees 2, 3, and 5. The mean values were almost identical in trees 1 and 4, but were somewhat variable between trees 2, 3, and 5. The

highest mean EVLD was measured in the control (healthy) zone at stem height 0.25 m in tree 3. Stem height had a similar effect on EVLD in trees 2 and 5, as well as in trees 1 and 4, resulting in the highest mean values at height 1.25 in the first two and 1.75 in the second two, respectively.

In addition, the EVLD in the wounded zone varied with height between trees. In majority of trees, the lowest mean EVLD was measured at a stem height of 0.75 m (trees 1, 2, 4, and 5), though at only stem height 1.75 in tree 3. The mean EVLD showed the highest values in the wounded zones of trees 1, 3, and 4 at stem height 0.25 m (trees 1, 3, and 4) and stem height 1.75 m (trees 2 and 5). Stem height 1.25 m was the least pronounced variable in the peak EVLD values of all stem heights in all trees.

Similar trends in EVLD with height were observed in both the control (healthy) and wounded zones, with a decrease at stem height 0.75 m (Figure 7). In general, the control (healthy) zone showed significantly higher values of EVLD in comparison to the wounded zone (Table 6). This can be observed in Figure 8b,c, where control (healthy) earlywood vessels are significantly wider in diameter and of more round/oval shape in comparison to vessels in wounded earlywood, especially those close to the wound.

Table 6. Results of repeated measures analysis of variance (ANOVA) for the narrow-leaved ash earlywood vessel lumen diameter (EVLD) in the control (healthy) and wounded zones from four different stem heights.

Property	Source	SS	d.f.	MS	F	p
EVLD (μm)	Tree $\times$ Zone	4281	4	1070	2.94	0.0365
	Stem height $\times$ Tree	21514	12	1793	9.1	0.0000
	Stem height $\times$ Zone	10429	3	3476	17.6	0.0000
	Stem height $\times$ Tree $\times$ Zone	35529	12	2961	15.0	0.0000

Notes: EVLD—earlywood vessel lumen diameter; SS—sum of squares; d.f.—degrees of freedom; MS—mean sum of squares.



Figure 8. Cross sections: (a) macroscopic image of the disk with visible wounds; (b) microscopic image of control (healthy) and (c) wounded narrow-leaved ash earlywood (sampled stem height 0.75 m given as an example), scale 200 μm .

The ANOVA results showed that tree-by-zone interaction was significant, indicating differences in the EVLD of the two zones in the whole tree. Highly significant interaction was determined in other three cases as well: for the stem height by tree, stem height by zone, and stem height by both tree and zone. However, the objective was to identify whether and to what extent the control (healthy) and wounded zone EVLDs differed with different stem heights. Thus, the following discussion focuses on the zone main effect and the stem height by zone interaction terms.

The effect of wood zone was significant for all stem heights except for 0.25 m (Table 7). The results indicate the expected significant differences in EVLD between the control (healthy) and wounded zones at greater stem heights.

Table 7. Results of repeated measures analysis of variance (ANOVA) for the narrow-leaved ash earlywood vessel lumen diameter (EVLD) in the control (healthy) and wounded zones by separately observing four different stem heights.

Property	Stem Height (m)	Source	SS	d.f.	MS	F	<i>p</i>
EVLD (μm)	0.25	Zone	12,883.8	1	12,883.8	4.8	0.0594
	0.75	Zone	42,432.8	1	42,432.8	21.2	0.0017
	1.25	Zone	49,862.0	1	49,862.0	71.8	0.0001
	1.75	Zone	61,930.6	1	61,930.0	12.0	0.0085

Notes: EVLD—earlywood vessel lumen diameter; SS—sum of squares; d.f.—degrees of freedom; MS—mean sum of squares.

4. Discussion

4.1. Bark Stripping Inventory

In our study, narrow-leaved ash was confirmed to be a highly food attractive species for red deer, as previously reported [54], with more than half (56%) of observed trees displaying bark stripping wounds. This is in accordance with similar studies conducted on common ash [18,40,55]. In comparison, the share of wounded trees for other observed species (elm, oak, alder, and pear) did not exceed 10%. This can be partially explained by the fact that this study was conducted in single-species forest stands where narrow-leaved ash was by far the most abundant and most available tree species, making it more liable to bark stripping. Moreover, it is possible that the thickness or composition of ash bark is more appealing to red deer in comparison to the bark of other trees species, which can be more robust or contain more tannins, although this needs to be further investigated.

The ratio of wounded and unwounded ash trees differed significantly among the three studied locations, as did their average DBH, indicating that these parameters are associated. The incidence of wounded trees increased with the decrease in average DBH. In location L1, where the average DBH was 10 cm, 46% of ash trees were wounded; in location L2, the average DBH was 7.5 cm, and the share of wounded trees was 54%; and in location L3, where the average DBH was 5 cm, the number of wounded trees (81%) was four times the number of unwounded trees. It can be concluded that younger ash trees, especially those with a DBH around 5 cm, were the most appealing to red deer in the researched forest stands. This was additionally corroborated by the fact that the average DBH of all wounded ash trees from all three locations was 5 ± 2.5 cm. However, these variations in the number of wounded ash trees among locations were most probably affected by other factors as well. Several authors have pointed out that the frequency and intensity of bark stripping are impacted forest stand properties such as density, age, and horizontal and vertical composition [29,56], and are additionally influenced by the density of the red deer population [16]. Red deer tend to reside in younger forest stands with higher tree density and a thinner snow layer on the ground, as these conditions provide them with shelter and more food sources [29].

In this research it was determined that bark stripping stops when ash trees reach a certain age, specifically, a DBH of 18 cm. The reason for this is primarily because the bark of young trees is more digestible [25,57], as well as the fact that at a DBH greater than 18 cm the ash bark becomes too rough and too thick for red deer to strip off with their teeth.

Most of the bark stripping wounds occurred in the browsing zone at stem heights between 0.1 m and 1.9 m (average bark stripping height = 0.6 ± 0.4 m). These results are partially in concordance with results from a study on common ash, where 80% of all bark stripping damage occurred between 1.0 m and 1.9 m of height [18], in which the bark stripping was inflicted by both red deer and moose. Because moose are significantly larger and taller than red deer, they consequently inflict the most damage on higher parts of stem, which is why their results showed somewhat higher values compared to the results of our study.

Usually, older trees with greater DBH display more and larger bark stripping wounds because of the greater surface available for debarking and repeated wounding of the same trees over several years [54]. Although the bark stripping intensity (the share of damaged circumference) did increase slightly with the increase of DBH in this research, no significant association between these parameters was found, most probably because the circumference of all observed trees was relatively small and had a very narrow range. A positive association was found between the DBH of wounded trees and the height of bark stripping on stems (the lowest and highest points), most probably because the height of the trees increased with the DBH and because of the thinner bark on higher stem parts.

4.2. Fungal Presence in Bark Stripping Wounds

Fortunately, narrow-leaved ash seems to heal inflicted wounds relatively successfully and quickly, which has been reported for common ash as well [18]. Out of ten randomly selected wounded stems used for fungal isolation in this study, six displayed fully closed wounds where discoloration and decay were restricted to the annual rings that grew before the stripping, as described for European beech by other authors [58]. Three samples displayed freshly inflicted wounds, and only one represented a partially closed wound. This is probably related to tree vigor and associated intensified cambium growth, which are at their peak in younger trees, enabling faster wound healing [18,19]. Nevertheless, the further spread of discoloration and decay are possible with tree aging and loss of vitality, as certain fungal species continue to dwell in wooden tissue and advance even after wound closure [32]. Moreover, because bark-stripping causes physiological stress [18] and activates the allocation of resources to healing [15], tree growth is reduced and it is possible that less carbon is invested into the root system [59], making trees more prone to the effects of other abiotic and biotic stresses. A relationship between tree mortality caused by pathogenic fungi and previous damage to the trunk caused by deer has been observed in aspen (*Populus tremuloides* Michx.) [60].

Most of the fungi isolated from the wounded wooden tissue in this research were ubiquitous generalists and known saprotrophs or endophytes, similar to in the young bark stripping wounds on *Quercus robur* in Lithuania studied in [32]. Several of these species and genera (*Clonostachys rosea*, *Fusarium* spp., *Trichoderma* spp.) were reported to be endophytic in narrow-leaved ash in a previous study [9], indicating that they might have been present in the ash wood prior to bark stripping. However, further research is required to determine the exact colonization process and patterns of the fungi found in this study. Relatively young trees and wound ages might be responsible for the low incidence of several potentially pathogenic and decay fungi found in this research, as reported by Burneviča et al. [61] and Marčiulynas et al. [32]. Only two white-rot fungi were isolated, *Bjerkandera adusta* and *Hyphodermella rosae*, with rather low incidence; both have already been reported on common ash [12,62]. Restricted spread of decay following bark stripping has been reported for common ash as well [18]. This indirectly corroborates the finding that summer bark stripping leads to a greater risk of decay development in comparison to winter wounding [37]. Similarly, only two designated woody plant pathogens were

isolated: *Peroneutypa scoparia* (syn. *Eutypella scoparia*) and *Eutypa lata*. The first is associated with canker, stem blight, and dieback symptoms on blueberry, grapevine, and fig [63], while the latter is considered to be a pathogen with a wide host range, and is particularly fatal for grapevine and apricot [64]. Fungi of the genus *Eutypa* are quite abundant in silver fir trees (*Abies alba* Mill.) damaged by bark stripping, although mostly in older wounds [65]. While these pathogens were relatively abundant regarding the number of isolates obtained in this study, each occurred in only one sampled stem. The most frequently isolated taxa which colonized more than two stems were *Boeremia* spp., *Clonostachys rosea*, *Fusarium* spp., and *Trichoderma* spp., all of which are reported to be endophytes and potential opportunistic pathogens of ash. *Boeremia* isolates were found to be the most abundant fungi in necrotic common ash lenticels in [66], with the species *B. lilacis* and *B. exigua* reported to be endophytes with an ability to become pathogenic and cause minor damage to the host [8,66,67]. *Clonostachys rosea* has been isolated from both healthy and necrotic wooden tissues of common and narrow-leaved ash in previous studies [9,68], confirming its ability to spread in symptomatic tissue. Members of the genus *Fusarium* are widespread and colonize a great number of plant hosts, often without causing severe disease; these include *F. acuminatum* [69], which was the only *Fusarium* species identified to the species level in this study. This species has been found in old bark stripping wounds on silver fir as well [65]. *Fusarium* species have a tendency to become pathogenic, especially in stressed hosts, which has been confirmed in inoculation experiments on common ash, with three species able to cause necrotic lesions on the stems of some seedlings [8]. *Fusarium* species have been isolated from healthy and necrotic tissues of narrow-leaved ash as well [9]. The most frequently isolated fungi in this study were members of genus *Trichoderma*, which are known to be ubiquitous saprotrophs and endophytes in plants [70] and in certain cases antagonists of fungal pathogens [71]. Their abundance in narrow-leaved ash stems of different health status has already been reported [9], as has their presence in freshly inflicted and healed bark stripping wounds on silver fir [65]. Without further identification to a species level, it is hard to conclude whether they act antagonistically to other fungi and prevent the further spread of discoloration and decay or opportunistically contribute to the advantage of the mentioned symptoms.

4.3. Wood Structure Analysis

Spread of discoloration, decay, and associated microorganisms can be mitigated or completely stopped by tree defense mechanisms, which often lead to changes in the wood structure. No recent studies on bark stripping have dealt with its effect on vessel dimensions, which represent the main element of the xylem vascular tissue. Earlier studies [72,73] quantified changes in cell dimensions due to different tissue composition after wounding, and a somewhat similar topic was elaborated in [74] by artificially wounding hardwood trees at different stem heights.

Macroscopic observation of disks taken at four different stem heights allows for the determination of visible anomalies in the wood tissue based on the presence of certain wood defects. The inflicted damage can vary from minor scarring to significant deformations of the trunk [27]. It can be concluded that trees respond to wounds anatomically in order to isolate the wounded area [75,76]. In this study, discoloration, decay, partial detachment, and deformation of annual growth rings were observed on cross-sections at stem heights 0.25–1.25 m; at stem height 1.75 m, mostly bright covered knots were seen, as well as some defects similar to those seen at lower stem heights. There were zones that were impossible to sample due to the difficulty of cutting thin cross-sections for anatomical analysis, e.g., large zones containing cracks and decay due to excessively damaged xylem wood tissue. For this reason, the most optimal or most pronounced wounds may not have been selected for analysis.

The literature reports that the timing of stripping is mostly in the late winter and early spring, when the sap flow of trees is increasing [27]. It was expected that selected wounded earlywood zones from each of the selected stem heights would have vessels with different

dimensions. The main preliminary results of this study indicate significant differences in earlywood vessel lumen diameter between the control (healthy) and wounded zones in all sampled trees within the locations. Significant tree-by-zone and stem height-by-zone interactions were determined. Similar results for the first interaction have been confirmed by Lowerts et al. [74]. However, the same authors identified non-significant stem height-by-zone interaction, which is not in accordance with the results of this study. Welch and Scott [77] reported on small wounds that healed well without any significant damage present. Despite the determined differences in EVLD between the two zones, the size of the wounds selected in this study can be considered as the key factor, especially at stem height 0.25 m. This is due to insignificant differences between the two zones at the lowest height. If the stem height effect was individually investigated, no differences between the control (healthy) and wounded zone would be taken into consideration, whereas the zone effect was significant when analyzed separately and after including other interacting factors, i.e., tree and stem height. The high statistical significance of all individual factors contributed to the significance of their interactions.

Vessel lumen diameter is considered a key parameter of vessel dimension [78]. Vessel dimensions generally varies with stem height. However, the vertical variations of this wood anatomical characteristic have not been extensively investigated. One study on several tree species, including narrow-leaved ash, reported that vessel lumen diameter decreased with axial height, as the conduits gradually widen when moving downwards, which is a necessary anatomical feature for stabilizing hydrodynamic resistance with tree height [79,80]. The results of this study, which are suitable for comparison concerning the control (healthy) zone, do not follow this previously explained vertical trend. The EVLD decreased between stem heights of 0.25 m and 0.75 m in both the control (healthy) and wounded zones, after which it increased until the highest stem height. In addition, higher EVLD values in the control (healthy) zone at all stem heights indicate the presence of damage in wood in the range of the smallest and largest measured stem heights. These values are closer to the values reported by Wagenführ [45]. It is well known that the anatomical structure of wood is genetically determined [46]. The absence of a difference in EVLD distribution with the height of both the zones, as seen in Figure 7, could be associated with the small distance between the analyzed zones within the disks or with some other environmental effect on trees.

The characteristics of the wounded wood were similar to those of other hardwoods described in the literature. As described by Tippet and Shigo [81] and Luley [82], xylem cells produced by the vascular cambium form a specific zone following wounding, known as the “barrier zone”. The cells in this distinct zone differ, mainly visually, from the cells in healthy zones. The “barrier zone” is observed as a band of parenchyma, as described by Lowerts et al. [74], with the earlywood vessels in the proximity being significantly smaller. Microscopic images of wounded zone cross-sections (Figure 8c) show that the wounded zone has been effectively healed by phenolic substances in the medullary rays (seen as vertical lines). The separation of the wood as a fracture, as seen in Figure 8c, occurs between of the growth layers, commonly together with discoloration. A similar visual anomaly in the wood tissue called “ring shake”, in which internal fractures depend on the balance between stresses and strength, was described by Fonti and Macchioni [83]. The earlywood zone is generally an area of structural weakness, and is especially vulnerable to stresses which can cause separation [83]. Nevertheless, it is not easy to make definite conclusions about the causes of such fractures, and ring shakes are not the subject of this study. Mechanical damage to the tree is more likely to be the explanation for the poor bonding quality and deterioration of the observed wounded zones.

It is common for tyloses to be present in the earlywood vessels of narrow-leaved ash wood [45]. However, the analyzed healthy wood tissues did not contain tyloses of any kind, nor were the vessels observed in the wounded zone broken, and no vessel occlusions with tyloses were present. The only substantial difference was that they were narrower in terms of lumen diameter. At the time of the damage, water conductivity was likely affected

by the development of smaller vessels. In accordance with this, the decrease in longitudinal and lateral conductivity would serve to inhibit the spread of microorganisms [74].

Bark stripping was observed on ash trees from all three locations from 2 cm up to 18 cm of DBH and between 0.1 m and 1.9 m stem height. However, no significant differences were found between control (healthy) and wounded zones at a stem height of 0.25 m, only at the other three stem heights. These results point to the possibility that the size of the wounds was not optimal for interpretation. Considering that only a small proportion of damage to the wood structure was analyzed, these results cannot be generalized as much as if all possible effects were analyzed.

5. Conclusions

Narrow-leaved ash can be confirmed as a highly attractive food source for red deer during winter months, as more than half of observed trees were wounded. Younger trees with smaller DBH and thinner bark were more susceptible to wounding, particularly those with average DBH 5 ± 2.5 cm. The share of wounded trees increased with decreasing average DBH in all three studied locations. When the trees reached a certain age, i.e., a DBH of 18 cm, the bark stripping stopped. The height of bark stripping wounds on a stem increased with increasing DBH as well, most likely due to the consequential increase in overall height of older ash trees and the fact that higher stem parts carry thinner bark.

Young narrow-leaved ash trees have been shown to be fast healers of wounds inflicted by bark stripping. On most of the samples the wounds were completely closed and discoloration and decay were limited to the central part of stem, i.e., the wooden tissue present before the injury has happened. The wounds were mostly inhabited by widespread generalist fungi, mostly *Trichoderma* spp., *Fusarium* spp., and *Clonostachys rosea*, which are reported to be endophytes, saprotrophs, or potential opportunistic pathogens of plants. Decay and pathogenic fungi were present but rare. Although most of these fungi are likely to have been present in the narrow-leaved ash wood prior to wounding, as they are reported to be endophytic in this species, further research is required to study their colonization patterns in bark stripping wounds in narrow-leaved ash.

Our preliminary results confirm the effect of bark stripping on the wood structure of narrow-leaved ash, indicating significantly narrower vessel lumens within the earlywood wounded zone. The abnormal wood tissue in the wounded zone most likely has a mechanical origin. The results of our wood anatomical analysis were limited to smaller damage due to methodology constraints, specifically the need to avoid complete separation and damage to the wood tissue while cutting the samples. Further analysis should include wood mechanical properties in order to better evaluate wood quality after bark stripping damage by measuring and analyzing larger wound areas.

Despite being a fast healer of wounds at an early age, narrow-leaved ash has proven to be very susceptible to bark-stripping by red deer at same stage when it is more susceptible to *H. fraxineus* infections. Although this pathogen was not found in the current study, its presence in wounded trees cannot be excluded and might be confirmed by increased sampling efforts. Wounds were found to be colonized by other pathogenic and decay fungi as well as by endophytes, which could potentially contribute to the process of ash dieback.

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